

Mathematics for Data Science

Lecture 3 - Determinant

Eva FEILLET¹

¹LISN

Université Paris-Saclay, CNRS,
91190 Gif-sur-Yvette, France

M1, Fall 2026

Previously covered topics

- (Lecture 1) Vector spaces, subspaces, linear transformations. Rank, image, kernel
- (Lecture 2) Matrices, link with linear transformations, linear systems. Range, rank, kernel of a matrix, inverse of a matrix

Previously covered topics

- (Lecture 1) Vector spaces, subspaces, linear transformations. Rank, image, kernel
- (Lecture 2) Matrices, link with linear transformations, linear systems. Range, rank, kernel of a matrix, inverse of a matrix

In this lecture: determinant and its properties, beginning of diagonalization

Introduction

Determinants were first introduced in the context of systems of linear equations with as many unknowns as equations, in order to **determine** whether a system admitted a unique solution.

Works by Cardan (2 equations with 2 unknowns, 1545), Leibniz (3 by 3, 1678), Maclaurin (4 by 4, 1748), Cramer (formula for n by n but no proof), Bézout, Lagrange, Gauss, Cauchy...



Figure 1: Gerolamo Cardano, Italian mathematician from the XVIth century

Determinant: definitions

The **determinant** of a square matrix A , denoted $\det(A)$, or $|A|$, can be defined in several different ways.

Determinant: definitions

The **determinant** of a square matrix A , denoted $\det(A)$, or $|A|$, can be defined in several different ways.

- ① It can be defined by the Leibniz formula, involving permutations and their signatures.
- ② In Euclidean spaces, its absolute value can be interpreted in terms of area and volume, and its sign in terms of orientation.
- ③ It can be characterized as the unique function, defined on the entries of a matrix, satisfying a given set of properties (the unique alternating multilinear form normalized on a basis).

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Determinant: explicit formulae

The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Determinant: explicit formulae

The determinant of a 2×2 matrix is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and the determinant of a 3×3 matrix is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Example

On the board

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Geometrical Interpretation in 2D

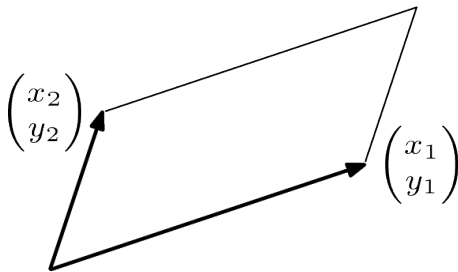


Figure 2: Two vectors in $\mathbb{R}^2$.

Geometrical Interpretation in 2D

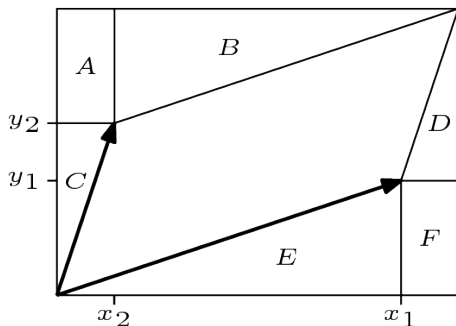
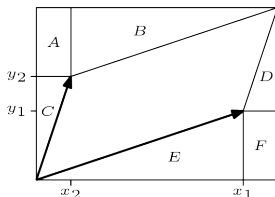


Figure 3: Computing the area of the parallelogram defined by the two vectors in $\mathbb{R}^2$. [Image source](#).

Geometrical Interpretation in 2D



$$\begin{aligned} & \text{area of parallelogram} \\ &= \text{area of rectangle} - \text{area of } A - \text{area of } B \\ & \quad - \dots - \text{area of } F \\ &= (x_1 + x_2)(y_1 + y_2) - x_2 y_1 - x_1 y_1 / 2 \\ & \quad - x_2 y_2 / 2 - x_2 y_2 / 2 - x_1 y_1 / 2 - x_2 y_1 \\ &= x_1 y_2 - x_2 y_1 \end{aligned}$$

Figure 4: Computing the area of the parallelogram defined by the two vectors in $\mathbb{R}^2$. [Image source](#).

Geometrical Interpretation in 2D: Orientation matters !

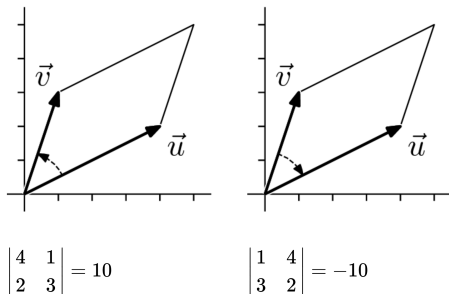


Figure 5: Computing the area of the parallelogram defined by the two vectors in $\mathbb{R}^2$. Orientation matters ! Oriented surfaces. [Image source](#).

Proposition

- 1 Exchanging two columns flips the sign of the determinant.
- 2 Scaling a vector by a factor λ multiplies the determinant by λ .

Geometrical Interpretation in 2D - Area

Consider

$$v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad M = [v|w] = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ -2 & 0 \end{bmatrix}$$

In $\mathbb{R}^2$, the transformation of the parallelogram by t , with matrix A , scales the area by a factor of $|\det(A)|$.

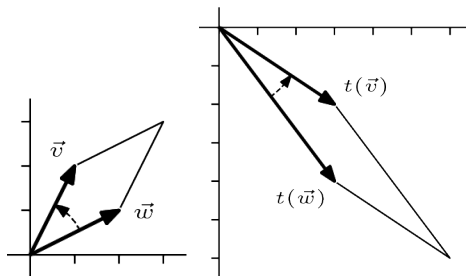


Figure 6: Example with a linear transformation [Image source](#).

Geometrical Interpretation in 3D

Volume of a parallelepiped:

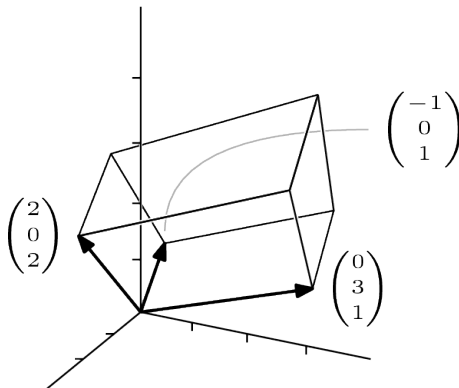


Figure 7: The volume of the box defined by vectors v_1, v_2, v_3 is the absolute value of the determinant of the matrix with v_1, v_2, v_3 as columns. [Image source](#).

◇ Practice with [exercises](#) / [solutions](#)

ML illustration: determinant as volume scaling

Let $A : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a linear transformation. A unit ball is transformed into a hyper-ellipsoid via multiplication by A .

The determinant measures how A changes volumes: volume is scaled by $|\det(A)|$.

$$\text{new volume} = |\det(A)| \times \text{original volume}$$

- If $|\det(A)| > 1$, volumes are expanded.
- If $0 < |\det(A)| < 1$, volumes are contracted.
- If $\det(A) = 0$, some directions are collapsed.

Interpretation : In ML, linear feature transformations can stretch, compress, or collapse regions of the feature space.

ML illustration: determinant in Gaussian models

For a multivariate Gaussian distribution

$$x \sim \mathcal{N}(\mu, \Sigma),$$

the density contains the determinant of the covariance matrix:

$$p(x) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right).$$

Interpretation: $\det(\Sigma)$ measures the volume of the uncertainty ellipsoid.

- Large $\det(\Sigma)$: spread-out distribution.
- Small $\det(\Sigma)$: concentrated distribution.

Properties in 2D: Flipping and scaling

Proposition

- 1 Exchanging two columns, or two rows, flips the sign of the determinant.
- 2 Scaling one column, or one row, by a factor λ multiplies the determinant by λ .
- 3 The determinant is linear with respect to one column, or one row, given that the others are fixed.

Properties in 2D: Flipping and scaling

Example

Flipping columns :

$$\begin{vmatrix} b & a \\ d & c \end{vmatrix} = (bc - ad) = - \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Scaling one column :

$$\begin{vmatrix} \lambda a & b \\ \lambda c & d \end{vmatrix} = \lambda(ad - bc) = \lambda \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Adding a vector to the first row :

$$\begin{vmatrix} a + a' & b + b' \\ c & d \end{vmatrix} = (ad - bc) + (a'd - b'c) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

Table of Contents

1 Determinant

- Explicit formulae
- Geometrical interpretation
- Signature of a permutation
- Multilinear, alternating map
- Determinant and invertibility
- Determinant of an endomorphism

2 Determinant of some particular matrices

3 Eigenvalues and eigenvectors

Why introduce permutations?

For 2×2 and 3×3 matrices, explicit formulas are manageable.

For an $n \times n$ matrix, the determinant is a *signed sum over all ways of choosing one coefficient in each row and each column.*

These choices are encoded by permutations.

Permutation and transposition

Definition (Permutation)

Let $n \in \mathbb{N}^*$. We denote by $\mathcal{S}_n$ the set of permutations in $\llbracket 1, n \rrbracket$.

NB : it corresponds to the bijections from $\llbracket 1, n \rrbracket$ to $\llbracket 1, n \rrbracket$. What is the cardinal of $\mathcal{S}_n$?

Permutation and transposition

Definition (Permutation)

Let $n \in \mathbb{N}^*$. We denote by $\mathcal{S}_n$ the set of permutations in $\llbracket 1, n \rrbracket$.

NB : it corresponds to the bijections from $\llbracket 1, n \rrbracket$ to $\llbracket 1, n \rrbracket$. What is the cardinal of $\mathcal{S}_n$? $\text{Card}(\mathcal{S}_n) : n!$.

Permutation and transposition

Definition (Permutation)

Let $n \in \mathbb{N}^*$. We denote by $\mathcal{S}_n$ the set of permutations in $\llbracket 1, n \rrbracket$.

NB : it corresponds to the bijections from $\llbracket 1, n \rrbracket$ to $\llbracket 1, n \rrbracket$. What is the cardinal of $\mathcal{S}_n$? $\text{Card}(\mathcal{S}_n) : n!$.

Definition (Transposition)

Let $(i, j) \in \llbracket 1, n \rrbracket, i \leq j$. We call **transposition**, and denote $\tau_{i,j}$ the map in $\mathcal{S}_n$ defined by: $\tau_{i,j} : \llbracket 1, n \rrbracket \rightarrow \llbracket 1, n \rrbracket$,

$$\tau_{i,j}(k) = k \text{ if } k \notin \{i, j\},$$

$$\tau_{i,j}(k) = i \text{ if } k = j,$$

$$\tau_{i,j}(k) = j \text{ if } k = i$$

NB : What is the inverse map of $\tau_{i,j}$?

Permutation and transposition

Definition (Permutation)

Let $n \in \mathbb{N}^*$. We denote by $\mathcal{S}_n$ the set of permutations in $\llbracket 1, n \rrbracket$.

NB : it corresponds to the bijections from $\llbracket 1, n \rrbracket$ to $\llbracket 1, n \rrbracket$. What is the cardinal of $\mathcal{S}_n$? $\text{Card}(\mathcal{S}_n) : n!$.

Definition (Transposition)

Let $(i, j) \in \llbracket 1, n \rrbracket, i \leq j$. We call **transposition**, and denote $\tau_{i,j}$ the map in $\mathcal{S}_n$ defined by: $\tau_{i,j} : \llbracket 1, n \rrbracket \rightarrow \llbracket 1, n \rrbracket$,

$$\begin{aligned}\tau_{i,j}(k) &= k \text{ if } k \notin \{i, j\}, \\ \tau_{i,j}(k) &= i \text{ if } k = j, \\ \tau_{i,j}(k) &= j \text{ if } k = i\end{aligned}$$

NB : What is the inverse map of $\tau_{i,j}$? $\tau_{i,j} \circ \tau_{i,j} = \text{Id}$. A transposition is its own inverse.

If $i = j$, then $\tau_{i,j} = \text{Id}$.

Permutation and transposition

Composition of transpositions

Any permutation $\sigma \in \mathcal{S}_n$ can be written as a composition of transpositions, i.e. for any permutation $\sigma \in \mathcal{S}_n$, there exists $\tau_1, \tau_2, \dots, \tau_T$, T transpositions in $\mathcal{S}_n$ such that

$$\tau_1 \circ \tau_2 \circ \dots \circ \tau_T = \sigma$$

NB : Proof by induction over n . Case $n = 1$: $\mathcal{S}_1 = \{Id\}$. Case $n = 2$: $\mathcal{S}_2 = \{Id, \tau_{1,2}\}$. Case $n = 3 \dots$

Permutation and transposition

Definition (Signature of a permutation)

Let $\sigma \in \mathcal{S}_n$. We call **signature** of a permutation σ the real number, denoted $\epsilon(\sigma)$ defined by:

$$\epsilon(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}$$

NB : what values can the signature take ?

Permutation and transposition

Definition (Signature of a permutation)

Let $\sigma \in \mathcal{S}_n$. We call **signature** of a permutation σ the real number, denoted $\epsilon(\sigma)$ defined by:

$$\epsilon(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}$$

NB : what values can the signature take ? $\epsilon(\sigma) \in \{-1, 1\}$
what about compositions of signatures ?

Permutation and transposition

Definition (Signature of a permutation)

Let $\sigma \in \mathcal{S}_n$. We call **signature** of a permutation σ the real number, denoted $\epsilon(\sigma)$ defined by:

$$\epsilon(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}$$

NB : what values can the signature take ? $\epsilon(\sigma) \in \{-1, 1\}$

what about compositions of signatures ? $\epsilon(\sigma \circ \sigma') = \epsilon(\sigma)\epsilon(\sigma')$

Exercise : prove it !

Proposition

The signature of a transposition is -1 .

Signature of a permutation / Recap

- Definition of a **permutation** σ of the elements of a finite set X
 - $\mathcal{S}_n$ the set of permutations of integers in $[1, n]$.
 - $\text{Card}(\mathcal{S}_n) = n!$.
- Definition of a **transposition** $\tau_{i,j}$
- the signature of a permutation can be defined from its decomposition into a product of T transpositions: $\epsilon(\sigma) = (-1)^T$
- An **inversion** is a pair (i, j) such that $i < j$ and $\sigma(i) > \sigma(j)$.
- alternatively, we can write the signature of a permutation as : $\epsilon(\sigma) = (-1)^{N(\sigma)}$ where $N(\sigma)$ is the number of inversions in σ

Signature of a permutation / Recap

- Definition of a **permutation** σ of the elements of a finite set X
 - S_n the set of permutations of integers in $[1, n]$.
 - $\text{Card}(S_n) = n!$.
- Definition of a **transposition** $\tau_{i,j}$
- the signature of a permutation can be defined from its decomposition into a product of T transpositions: $\epsilon(\sigma) = (-1)^T$
- An **inversion** is a pair (i, j) such that $i < j$ and $\sigma(i) > \sigma(j)$.
- alternatively, we can write the signature of a permutation as : $\epsilon(\sigma) = (-1)^{N(\sigma)}$ where $N(\sigma)$ is the number of inversions in σ

But we wanted to talk about determinants ?!

Signature of a permutation / Recap

- Definition of a **permutation** σ of the elements of a finite set X
 - S_n the set of permutations of integers in $[1, n]$.
 - $\text{Card}(S_n) = n!$.
- Definition of a **transposition** $\tau_{i,j}$
- the signature of a permutation can be defined from its decomposition into a product of T transpositions: $\epsilon(\sigma) = (-1)^T$
- An **inversion** is a pair (i, j) such that $i < j$ and $\sigma(i) > \sigma(j)$.
- alternatively, we can write the signature of a permutation as : $\epsilon(\sigma) = (-1)^{N(\sigma)}$ where $N(\sigma)$ is the number of inversions in σ

But we wanted to talk about determinants ?!

Definition (Determinant / version with signature of permutations)

Let $A = (a_{ij})$ be a $n \times n$ matrix. We define its **determinant** as:

$$\det(A) = \sum_{\sigma \in S_n} \epsilon(\sigma) \prod_{j=1}^n a_{\sigma(j), j}.$$

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - **Multilinear, alternating map**
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Multilinear, alternating map

Definition (Multilinear form)

A **multilinear form** on a vector space E over a field $\mathbb{K}$ is a map $f: E^n \rightarrow \mathbb{K}$ that is separately $\mathbb{K}$ -linear in each of its n arguments, i.e. for each $i \in \llbracket 1, n \rrbracket$ and $(u_1, u_2, \dots, u_n) \in E^n$, the map $f_i: E \rightarrow \mathbb{K}$ defined as follows is linear:

$$x \mapsto f(u_1, \dots, u_{i-1}, x, u_{i+1}, \dots, u_n)$$

Definition (Alternating form)

Let $f: E^n \rightarrow \mathbb{K}$ be a multilinear form. We say that f is **alternating** if, whenever two arguments are equal, the value of f is zero:

$$\forall (u_1, \dots, u_n) \in E^n,$$

$$[\exists (i, j) \in \llbracket 1, n \rrbracket^2, (i \neq j) \text{ and } u_i = u_j] \implies f(u_1, \dots, u_n) = 0]$$

Determinant: definition via properties

Proposition

Let E be a $\mathbb{K}$ -vector space of dimension n and $\mathcal{B} = (e_1, \dots, e_n)$ a basis of E . There exist a unique map $\det_{\mathcal{B}} : E^n \rightarrow \mathbb{K}$ such that

- 1 $\det_{\mathcal{B}}$ is multilinear
- 2 $\det_{\mathcal{B}}$ alternating, and
- 3 $\det_{\mathcal{B}} = 1$.

Determinant: definition via properties

Proposition

Let E be a $\mathbb{K}$ -vector space of dimension n and $\mathcal{B} = (e_1, \dots, e_n)$ a basis of E . There exist a unique map $\det_{\mathcal{B}} : E^n \rightarrow \mathbb{K}$ such that

- 1 $\det_{\mathcal{B}}$ is multilinear
- 2 $\det_{\mathcal{B}}$ alternating, and
- 3 $\det_{\mathcal{B}} = 1$.

Link with the determinant of a matrix A :

$$\det(A) = \det_{\mathcal{B}}(C_1, C_2, \dots, C_n)$$

where $C_1, \dots, C_n$ are the columns of A written in the basis $\mathcal{B}$.

Determinant: definition via properties / Recap

Proposition

- 1 The determinant of the identity matrix is 1: $\det(I_n) = 1$.
- 2 The exchange of two rows (or cols.) multiplies the determinant by -1 .
- 3 Multiplying one row (or one column) by a scalar λ multiplies the determinant by λ .
- 4 Adding a multiple of one row (resp. column) to another row (resp. column) does not change the determinant.

Laplace expansion (cofactor expansion)

Notation. For $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{K})$, let M_{ij} be the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j from A . The cofactor is $C_{ij} = (-1)^{i+j} \det(M_{ij})$.

Developing along column j (fixed j)

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij} = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}).$$

Developing along row i (fixed i)

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}).$$

Properties of the determinant

Proposition

Let A, B be square matrices of size $n \times n$.

- $\det(I_n) = 1$
- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \det(B)$ i.e. The determinant of a product of matrices is the product of their determinants.
- $\det(\alpha A) = \alpha^n \det(A)$
- The determinant is linear with respect to one row, given the other rows are fixed.

Example

Exercise session: use these properties to compute determinants.

Table of Contents

1 Determinant

- Explicit formulae
- Geometrical interpretation
- Signature of a permutation
- Multilinear, alternating map
- Determinant and invertibility
- Determinant of an endomorphism

2 Determinant of some particular matrices

3 Eigenvalues and eigenvectors

Determinant and invertibility

Basis change and determinant

Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of a vector space E of dimension n . Let P be the change-of-basis matrix whose j -th column is $[e'_j]_{\mathcal{B}}$, where $\mathcal{B}' = (e'_1, \dots, e'_n)$. Thus

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$$

Then, for every $(v_1, \dots, v_n) \in E^n$,

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(P) \det_{\mathcal{B}'}(v_1, \dots, v_n).$$

Equivalently,

$$\det_{\mathcal{B}'}(v_1, \dots, v_n) = \frac{1}{\det(P)} \det_{\mathcal{B}}(v_1, \dots, v_n).$$

Determinant and invertibility

Proposition

Let $\mathcal{B}$ be a basis of E , and $(v_1, v_2, \dots, v_n) \in E^n$. Then $(v_1, v_2, \dots, v_n)$ is a basis of E if and only if $\det_{\mathcal{B}}(v_1, v_2, \dots, v_n) \neq 0$.

Proof : $\implies$ then $\Longleftarrow$ by contradiction

Determinant and invertibility

Reminder: if A is invertible, it corresponds to an isomorphism.

Proposition

$$(A \text{ is } \mathbf{invertible}) \Leftrightarrow \det(A) \neq 0$$

$$(A \text{ is } \mathbf{singular}) \Leftrightarrow \det(A) = 0$$

Proposition

If A is invertible:

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

NB: We have

$$\det(AA^{-1}) = \det(A) \det(A^{-1}) \text{ and } \det(AA^{-1}) = \det(I_n) = 1$$

ML illustration: determinant and information loss

Consider a linear feature transformation

$$z = Ax, \quad A \in \mathbb{R}^{d \times d}.$$

If

$$\det(A) \neq 0,$$

then A is invertible: the original feature vector can be recovered from z .

If

$$\det(A) = 0,$$

then A collapses at least one direction, and some information is lost.

Interpretation: the determinant detects whether a square linear representation preserves or destroys information.

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Reminder : Theorem of basis change

Theorem (Theorem of basis change)

Let $\mathcal{B} = (u_1, u_2, \dots, u_n)$ and $\mathcal{B}' = (v_1, v_2, \dots, v_n)$ be two bases of a vector space E , $L : E \rightarrow E$ a linear map, $A = [L]_{\mathcal{B}}$ the matrix of L in $\mathcal{B}$ and $B = [L]_{\mathcal{B}'}$ the matrix of L in $\mathcal{B}'$.

Reminder : Theorem of basis change

Theorem (Theorem of basis change)

Let $\mathcal{B} = (u_1, u_2, \dots, u_n)$ and $\mathcal{B}' = (v_1, v_2, \dots, v_n)$ be two bases of a vector space E , $L : E \rightarrow E$ a linear map, $A = [L]_{\mathcal{B}}$ the matrix of L in $\mathcal{B}$ and $B = [L]_{\mathcal{B}'}$ the matrix of L in $\mathcal{B}'$.

Let P be the matrix such that the j^{th} column is $[v_j]_{\mathcal{B}}$, the coordinates of basis vector v_j of $\mathcal{B}'$ in the basis $\mathcal{B}$.

$$P = [[v_1]_{\mathcal{B}} \dots [v_n]_{\mathcal{B}}], \text{ so that } [x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$$

Then P is invertible and we have

$$B = P^{-1}AP.$$

Determinant of an endomorphism

Proposition

The determinant is invariant under matrix similarity.

From the theorem of basis change, we write:

$$\begin{aligned}\det(B) &= \det(P^{-1}) \det(A) \det(P) = \det(A) \det(P) \det(P^{-1}) \\ &= \det(A) \det(PP^{-1}) = \det(A) \det(I_n) = \det(A)\end{aligned}$$

In other words, given a linear endomorphism of a finite-dimensional vector space, the determinant of the matrix that represents this linear endomorphism on a given basis does not depend on this chosen basis. **This allows defining the determinant of a linear endomorphism, which does not depend on the choice of a coordinate system.**

Determinant of an endomorphism

Proposition

Let f and g two linear applications from E to E . Then:

- f is bijective iff $\det(f) \neq 0$.
- $\det(f \circ g) = \det(f)\det(g)$
- If $\mathcal{B} = (v_1, \dots, v_n)$ is a basis of E , then

$$\det_{\mathcal{B}}(f(v_1), \dots, f(v_n)) = \det(f)\det_{\mathcal{B}}(v_1, \dots, v_n)$$

- If f is bijective, then $\det(f^{-1}) = \frac{1}{\det(f)}$.

NB : we can write $\det(f)$ without specifying the basis because it does not depend on the choice of basis (cf previous property).

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Determinant of a diagonal matrix

Proposition

The determinant of a diagonal matrix is the product of its diagonal entries.

$$\det(\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)) = \prod_{i=1}^n \lambda_i$$

Corollary

A diagonal matrix $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is invertible if and only if all its diagonal coefficients are nonzero.

Determinant of a triangular matrix

Proposition

The determinant of a triangular matrix is the product of its diagonal entries.

Exercise : show this property. Hint : proof by induction over n the size of the matrix.

Determinant of an orthogonal matrix

Reminder: Definition of an orthogonal matrix

Determinant of an orthogonal matrix

Reminder: Definition of an orthogonal matrix

If U is orthogonal, then $U^\top U = I_n$. So $U^{-1} = U^\top$. Taking determinants:

$$\det(U^\top) \det(U) = 1$$

Since $\det(U^\top) = \det(U)$, we have:

$$(\det(U))^2 = 1 \quad \Rightarrow \quad \det(U) = \pm 1$$

Proposition

The determinant of an orthogonal matrix $U \in \mathcal{M}_n(\mathbb{R})$ is either 1 or -1 .

NB: $\det(U) \in \{+1, -1\}$ means that orientation is preserved if $+1$, flipped if -1 .

Why the name “orthogonal” matrix

Reminder: Scalar product of two vectors of $\mathbb{R}^n$.

Proposition

Let us consider $U \in \mathbb{R}^{n \times n}$ an orthogonal matrix. Columns (and rows) of U form orthonormal families.

NB: see Lecture 5 on normed spaces. Reminder: (1) two vectors $u, v \in \mathbb{R}^n$ are orthogonal if $u \cdot v = 0$, where $u \cdot v = u^\top v$ is the usual scalar product on $\mathbb{R}^n$. (2) a vector is normalized if its norm is 1, i.e. $\|u\|_2 = \sqrt{u \cdot u} = 1$.

Classic example: rotation in $\mathbb{R}^2$

Example

For an angle $\theta \in \mathbb{R}$, we define the rotation matrix as

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

- $R(\theta)$ rotates every vector in $\mathbb{R}^2$ counterclockwise by θ .
- $\det R(\theta) = \cos^2 \theta + \sin^2 \theta = 1$ (so orientation is preserved).
- $R(0) = I_2$, $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$.

Classic example: rotation in $\mathbb{R}^2$

Example

For an angle $\theta \in \mathbb{R}$, we define the rotation matrix as

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

- $R(\theta)$ rotates every vector in $\mathbb{R}^2$ counterclockwise by θ .
- $\det R(\theta) = \cos^2 \theta + \sin^2 \theta = 1$ (so orientation is preserved).
- $R(0) = I_2$, $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$.

Actually useful in ML ! e.g. see RoPE paper.

Verifying $R(\theta)$ is orthogonal

Compute $R(\theta)^\top R(\theta)$:

$$R(\theta)^\top R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Verifying $R(\theta)$ is orthogonal

Compute $R(\theta)^\top R(\theta)$:

$$\begin{aligned} R(\theta)^\top R(\theta) &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{pmatrix} \end{aligned}$$

Verifying $R(\theta)$ is orthogonal

Compute $R(\theta)^\top R(\theta)$:

$$\begin{aligned} R(\theta)^\top R(\theta) &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2. \end{aligned}$$

Hence $R(\theta)^{-1} = R(\theta)^\top = R(-\theta)$.

Orthonormal columns (Lecture 5)

The columns of $R(\theta)$ are

$$u_1 = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad u_2 = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}.$$

We have

$$\|u_1\|_2^2 = \cos^2 \theta + \sin^2 \theta = 1, \quad \|u_2\|_2^2 = \sin^2 \theta + \cos^2 \theta = 1,$$

$$u_1^\top u_2 = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0.$$

Thus the columns of $R(\theta)$ form an orthonormal basis of $\mathbb{R}^2$.

Orthogonal matrix and norms (Lecture 5)

Proposition

Inner products and norms are preserved when applying a multiplication by an orthogonal matrix.

$$(Qx) \cdot (Qy) = x \cdot y, \quad \|Qx\|_2 = \|x\|_2.$$

Exercise : proof (Lecture 5)

Example (Numerical example ($\theta = \pi/6$))

$$R\left(\frac{\pi}{6}\right) = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}, \quad R\left(\frac{\pi}{6}\right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{pmatrix}.$$

$$\left\| R\left(\frac{\pi}{6}\right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\|_2 = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1 \quad (\text{norm preserved}).$$

Orthogonal matrices

Example

More examples of orthogonal matrices

- **Reflections** (determinant -1), e.g. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (reflection across the x -axis).
- **Permutation matrices**, e.g. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (swap coordinates).

Summary

- Q orthogonal $\iff Q^T Q = I \iff Q^{-1} = Q^T$.
- Orthogonal matrices preserve lengths, angles, and dot products.
- In $\mathbb{R}^2$, $R(\theta)$ is orthogonal and models rotation by θ .
- Orthogonal matrices have determinant ± 1 (rotations vs. reflections).

Table of Contents

- 1 Determinant
 - Explicit formulae
 - Geometrical interpretation
 - Signature of a permutation
 - Multilinear, alternating map
 - Determinant and invertibility
 - Determinant of an endomorphism
- 2 Determinant of some particular matrices
- 3 Eigenvalues and eigenvectors

Eigenvalues and eigenvectors

Definition

Let A be an $n \times n$ matrix. A nonzero vector $v \in \mathbb{R}^n$ is called an *eigenvector* of A if there exists $\lambda \in \mathbb{R}$ such that:

$$Av = \lambda v.$$

The scalar λ is called an *eigenvalue* of A associated with v .

Example

Find (i) eigenvalues of I_n and (ii) eigenvalues of $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

Eigenvalues and eigenvectors

Definition

Let A be an $n \times n$ matrix. A nonzero vector $v \in \mathbb{R}^n$ is called an *eigenvector* of A if there exists $\lambda \in \mathbb{R}$ such that:

$$Av = \lambda v.$$

The scalar λ is called an *eigenvalue* of A associated with v .

Example

Find (i) eigenvalues of I_n and (ii) eigenvalues of $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

- Any vector $x \in \mathbb{R}^n$ satisfies $I_n x = 1 \cdot x$, i.e. so every nonzero vector is an eigenvector of I_n associated with the eigenvalue 1.
- For $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, each canonical vector e_i is an eigenvector associated with the eigenvalue λ_i .

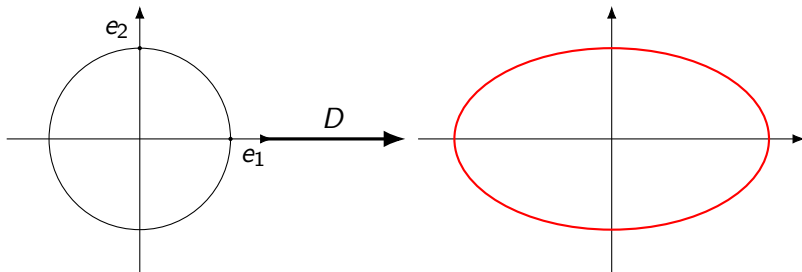
Geometrical interpretation

Example

In 2D : represent v an eigenvector of matrix A associated with eigenvalue λ . Discuss according to the value of λ .

In $\mathbb{R}^2$, how is the unit circle transformed via D ?

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{Unit circle } \left\{ x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mid x_1^2 + x_2^2 = 1 \right\}.$$



Geometrical interpretation

The circle is transformed into the red ellipse.

Let

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = Dx \quad \text{with } x \in \text{unit circle (i.e. } x_1^2 + x_2^2 = 1), \quad y = \begin{pmatrix} 2x_1 \\ x_2 \end{pmatrix}.$$

Hence $x_1 = \frac{y_1}{2}$, $x_2 = y_2$, and substituting into $x_1^2 + x_2^2 = 1$ gives

$$\boxed{\frac{y_1^2}{4} + y_2^2 = 1} \quad (\text{Equation of an ellipse}).$$

Eigenspace and eigenspectrum

Definition (Eigenspace)

The **eigenspace** of A associated with an eigenvalue λ is the set of all eigenvectors associated with λ , together with the zero vector:

$$E_\lambda = \{v \in \mathbb{R}^n \mid Av = \lambda v\} = \ker(A - \lambda I_n)$$

Exercise : prove that it is a subspace.

NB : If a set of eigenvectors of A forms a basis of the domain of A , then this basis is called an **eigenbasis**.

Definition (Eigenspectrum)

The **eigenspectrum** (or **spectrum**) of a matrix is the list of its eigenvalues, repeated according to their multiplicity.

NB: We will see that an important quantity associated with the spectrum is the maximum absolute value of any eigenvalue. This is known as the **spectral radius** of the matrix.

Diagonalization

Proposition

If A has n linearly independent eigenvectors $\{v_1, \dots, v_n\}$ with associated eigenvalues $\{\lambda_1, \dots, \lambda_n\}$, then these eigenvectors form a basis of $\mathbb{R}^n$ and A is **diagonalizable**:

$$A = VDV^{-1}$$

where $V = (v_1 \dots v_n)$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

NB: with the above notations, we can also write

$$AV = VD$$

Diagonalization

In other words, if $V \in \mathcal{M}_{n \times n}(\mathbb{R})$ is composed of n linearly independent vectors:

$$\text{span}(v_1, \dots, v_n) = \mathbb{R}^n \Rightarrow \text{rank}(V) = n \Rightarrow V \text{ is an invertible matrix.}$$

From $AV = VD$, if V is invertible, we get

$$AV = VD \iff AVV^{-1} = VDV^{-1} \iff A = V \underset{\text{Diagonal}}{D} V^{-1}.$$

We say that the matrix A is **diagonalizable**.

Computing AV in detail

$$AV = A(v_1 \cdots v_n) = (Av_1 \cdots Av_n) = (\lambda_1 v_1 \cdots \lambda_n v_n) = VD.$$

$$V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix},$$

$$VD = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = \begin{bmatrix} | & & | \\ \lambda_1 v_1 & \cdots & \lambda_n v_n \\ | & & | \end{bmatrix}.$$

$$(VD)_{ij} = \sum_{k=1}^n V_{ik} D_{kj} = V_{ij} \lambda_j, \quad (AV)_{ij} = (Av_j)_i = (\lambda_j v_j)_i = V_{ij} \lambda_j,$$

hence $AV = VD$.

Notes around $AV = VD$

Warning

$$VD \neq DV$$

- VD multiplies the columns of V by the diagonal entries d_i .
- DV multiplies the rows of V by the diagonal entries d_i .

Exercise: Powers of a matrix with a basis of eigenvectors

Example

Let $\{(\lambda_i, v_i)\}_{i=1}^n$ be eigenvalue–eigenvector pairs of a matrix $A \in \mathbb{R}^{n \times n}$ such that $\{v_1, \dots, v_n\}$ are linearly independent. Identify the eigenvalues and eigenvectors of $A^2, \dots, A^k$ for $k \in \mathbb{N}^*$. Are these matrices diagonalizable?

Exercise: Powers of a matrix with a basis of eigenvectors

Since $Av_i = \lambda_i v_i$ and the v_i 's form a basis, we have, by induction on $k > 0$,

$$A^k v_i = \lambda_i^k v_i \quad \text{for } i = 1, \dots, n, \quad k \in \mathbb{N}^*.$$

Hence for every $k \in \mathbb{N}$ the vectors $v_1, \dots, v_n$ are eigenvectors of A^k and the corresponding eigenvalues are $\lambda_1^k, \dots, \lambda_n^k$.

Because the same n eigenvectors remain linearly independent, A^k is diagonalizable: if $V = (v_1 \cdots v_n)$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, then

$$A = VDV^{-1} \implies A^k = VD^kV^{-1} = V \text{diag}(\lambda_1^k, \dots, \lambda_n^k) V^{-1}.$$

Remarks.

- If some $\lambda_i = 0$, then 0 is an eigenvalue of A^k ;
- for $k = 0$ we get $A^0 = I$ with eigenvalues all equal to 1 and the same eigenvectors v_i .

Next class

- Positive definite matrices, positive semi-definite matrices, Gram matrix
- Spectral theorem, Singular value decomposition...

Remark

Preparation for next class: Read Chapter 4 of the [MML book](#) (4.1 to 4.4 to review determinants and eigendecomposition)